

A counterexample to sparse Euclidean grids in every finite-dimensional Banach space

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Source problem

The source paper is Daniel J. Fresen, *Euclidean arrangements in Banach spaces*, arXiv:1309.5526. Problem 4 appears across pages 3–4 of the source PDF.

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The same dependence on $\|a\|_0$ can be achieved with the use of the Dvoretzky-Rogers factorization [4, 11, 39, 40, 43], but only on a subspace of proportional dimension. Note that in Corollary 3, logarithmic sparsity alone is not enough to guarantee bounded Euclidean distortion.

Problem 4 *Does there exist a universal constant $C > 0$ and a sequence $(\omega_n)_1^\infty$ with $\lim_{n \rightarrow \infty} \omega_n = \infty$ such that the following is true? For every $n \in \mathbb{N}$ and any real Banach*

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space $(X, \|\cdot\|)$ of dimension n , there is a basis $(e_i)_1^n$ for X such that for any $a \in \mathbb{R}^n$ with $\|a\|_0 \leq \omega_n$,

$$\left(\sum_{i=1}^n a_i^2 \right)^{1/2} \leq \left\| \sum_{i=1}^n a_i e_i \right\| \leq C \left(\sum_{i=1}^n a_i^2 \right)^{1/2}$$

Can one take $\omega_n = c \log n$? Can one take $(e_i)_1^n$ to be orthonormal with respect to the John ellipsoid of B_X ?

In transcription, Problem 4 asks whether there are a universal constant $C > 0$ and a sequence $\omega_n \rightarrow \infty$ such that every real n -dimensional Banach space X admits a basis $(e_j)_{j=1}^n$ satisfying

$$\|a\|_2 \leq \left\| \sum_{j=1}^n a_j e_j \right\|_X \leq C \|a\|_2$$

whenever a is ω_n -sparse.

Proof Intuition

It is enough to test the problem on $X = \ell_\infty^n$. A basis of ℓ_∞^n is an invertible matrix T , and the desired estimate says that T embeds every sparse coordinate Euclidean ball into the cube with distortion C .

The obstruction comes from signed vectors with an intermediate support size h . The upper estimate says that, in each row of T , only $O_C(h)$ coordinates can be as large as $1/\sqrt{h}$ when h is much smaller than the available sparsity k . On the other hand, the lower estimate for all signed vectors $\varepsilon 1_S/\sqrt{h}$ says that every signed h -set must contain a positive proportion of its signs inside one of these row-dependent high coordinate patterns. There are only $2n$ such signed patterns, each of size $O_C(h)$, and they cannot cover all signed h -sets once $h \rightarrow \infty$.

Counting lemma. *Fix constants $M, \eta > 0$. Let $h = h(n)$ be a sequence of positive integers such that $h \rightarrow \infty$ and $h \leq \sqrt{\log n}$. In the set of signed coordinates*

$$\Omega_n = \{1, \dots, n\} \times \{-1, 1\},$$

consider signed h -sets of the form

$$E(S, \varepsilon) = \{(j, \varepsilon_j) : j \in S\}, \quad |S| = h, \quad \varepsilon_j \in \{-1, 1\}.$$

Then, for all sufficiently large n , no collection of at most $2n$ subsets of Ω_n , each of cardinality at most Mh , can have the property that every signed h -set contains at least ηh points from one member of the collection.

Proof. The total number of signed h -sets is

$$\binom{n}{h} 2^h.$$

Fix a set $H \subseteq \Omega_n$ with $|H| \leq Mh$. The number of signed h -sets containing at least $q_0 = \lceil \eta h \rceil$ points of H is at most

$$\sum_{q=q_0}^h \binom{Mh}{q} \binom{n}{h-q} 2^{h-q}.$$

For large n we have $h \leq n/2$, and hence

$$\frac{\binom{n}{h-q}}{\binom{n}{h}} \leq \left(\frac{2h}{n}\right)^q.$$

Using also $\sum_q \binom{Mh}{q} \leq 2^{Mh}$, the fraction of all signed h -sets hit by this fixed H is at most

$$h 2^{Mh} \left(\frac{h}{n}\right)^{\eta h}.$$

Multiplying by $2n$, the fraction hit by any collection of at most $2n$ such sets is bounded by

$$2nh 2^{Mh} \left(\frac{h}{n}\right)^{\eta h}.$$

Since $h \rightarrow \infty$ and $h \leq \sqrt{\log n}$, the logarithm of this last quantity tends to $-\infty$. Thus it is eventually less than 1, so some signed h -set is not covered. $\square$

Counterexample. *Fresen's Problem 4 has a negative answer. More precisely, fix $C \geq 1$. There do not exist sequences $n_m \rightarrow \infty$, $k_m \rightarrow \infty$, and linear maps $T_m : \mathbb{R}^{n_m} \rightarrow \ell_\infty^{n_m}$ such that, for every k_m -sparse vector $a \in \mathbb{R}^{n_m}$,*

$$\|a\|_2 \leq \|T_m a\|_\infty \leq C\|a\|_2.$$

Consequently there is no universal constant C and no sequence $\omega_n \rightarrow \infty$ satisfying Problem 4, even for the spaces $X = \ell_\infty^n$.

Proof. Assume toward a contradiction that such C , n_m , k_m , and T_m exist. Suppress the index m and write $T = (r_{ij})_{i,j=1}^n$, with row vectors $r_i = (r_{ij})_{j=1}^n$. For every row i and every set $A \subseteq \{1, \dots, n\}$ with $|A| \leq k$, the upper estimate gives

$$\left(\sum_{j \in A} r_{ij}^2 \right)^{1/2} \leq C.$$

Indeed, the left-hand side is the operator norm of the i th coordinate functional restricted to the coordinate subspace supported on A .

Let

$$h = \left\lfloor \min \left\{ \frac{k}{64C^2}, \sqrt{\log n} \right\} \right\rfloor.$$

Since $k \rightarrow \infty$ and $n \rightarrow \infty$, we have $h \rightarrow \infty$ along a subsequence; we work on that subsequence. Put

$$\theta = \frac{1}{4\sqrt{h}}.$$

For each row i , let

$$A_i = \{j : |r_{ij}| \geq \theta\}.$$

We claim that $|A_i| \leq 16C^2h$ for every i . If $|A_i| \geq k$, then choosing k elements of A_i would give

$$\left(\sum_{j \in A} r_{ij}^2 \right)^{1/2} \geq \theta\sqrt{k} = \sqrt{\frac{k}{16h}} \geq 2C,$$

contradicting the upper estimate. Hence $|A_i| < k$. If $|A_i| > 16C^2h$, then applying the upper estimate to A_i itself gives

$$C \geq \left(\sum_{j \in A_i} r_{ij}^2 \right)^{1/2} > \theta\sqrt{16C^2h} = C,$$

again a contradiction.

For each row i and each sign $s \in \{-1, 1\}$, define a subset of signed coordinates

$$H_i^s = \{(j, \varepsilon) \in \Omega_n : s\varepsilon r_{ij} \geq \theta\}.$$

The previous paragraph gives $|H_i^s| \leq 16C^2h$.

Now take any signed h -set

$$E(S, \varepsilon) = \{(j, \varepsilon_j) : j \in S\}, \quad |S| = h,$$

and define $a_j = \varepsilon_j/\sqrt{h}$ for $j \in S$, with $a_j = 0$ otherwise. The vector a is h -sparse and $\|a\|_2 = 1$. By the lower estimate there are a row i and a sign $s \in \{-1, 1\}$ such that

$$\sum_{j \in S} s r_{ij} \varepsilon_j \geq \sqrt{h}.$$

Let

$$B = \{j \in S : s r_{ij} \varepsilon_j \geq \theta\}.$$

The contribution of $S \setminus B$ to the last sum is at most $h\theta = \sqrt{h}/4$, so the contribution of B is at least $3\sqrt{h}/4$. On the other hand, by Cauchy's inequality and the upper estimate on the coordinate set B ,

$$\sum_{j \in B} s r_{ij} \varepsilon_j \leq \left(\sum_{j \in B} r_{ij}^2 \right)^{1/2} |B|^{1/2} \leq C|B|^{1/2}.$$

Therefore

$$|B| \geq \frac{9}{16C^2}h.$$

Equivalently, every signed h -set contains at least ηh points from one of the $2n$ sets H_i^s , where $\eta = 9/(16C^2)$.

This contradicts the counting lemma with $M = 16C^2$. Hence no such sequence of maps exists.

Finally, if Problem 4 were true with a universal constant C and a sequence $\omega_n \rightarrow \infty$, then applying it to $X = \ell_\infty^n$ and taking T_n to be the matrix whose columns are the asserted basis vectors would give exactly such a forbidden sequence with $k_n = \lfloor \omega_n \rfloor$ along large n . This proves the claimed negative answer. $\square$

Verification notes

The proof is deterministic and uses only the source problem's two-sided sparse estimate. It does not require T to be invertible, so it applies a fortiori to matrices whose columns are bases of ℓ_∞^n .

A bounded novelty check was performed on 2026-06-14: the run indexes and local packets were searched for arXiv:1309.5526, "Euclidean grid", "ellinfy", sparse/logarithmic keywords, and row-polytope phrases. External exact-phrase searches for the source problem together with "ell_infy", "sparse Euclidean grid", and "omega_n" did not reveal a known answer. Novelty confidence is moderate pending human review.

Relation to previous local work

This packet strengthens the previous local logarithmic-ceiling packet, which only ruled out superlogarithmic sparsity in ℓ_∞^n . The counting argument above rules out every unbounded sparsity sequence in ℓ_∞^n for fixed distortion C .

For reviewer convenience, the previous partial packet is nested inside this packet at

`superseded_partial/1309.5526_linf_sparse_grid_logarithmic_ceiling/`.

It is retained as provenance, not as a separate current claim.

Appendix: superseded logarithmic-ceiling partial result

Before the full counterexample above was found, lane 1 isolated the following partial obstruction. It is mathematically useful because it identifies why the remaining frontier originally looked like $k \simeq c \log n$.

Superseded partial theorem. Let $T : \mathbb{R}^n \rightarrow \ell_\infty^n$ be linear, let $C \geq 1$, and let $1 \leq k \leq n$. Suppose that for every k -sparse vector $a \in \mathbb{R}^n$,

$$\|a\|_2 \leq \|Ta\|_\infty \leq C\|a\|_2.$$

Then

$$k \leq 2C^2 \log(2n).$$

Thus no affirmative answer to Problem 4 could have $\omega_n / \log n \rightarrow \infty$.

Proof of the superseded partial theorem. Let $r_i = (T_{ij})_{j=1}^n$ be the i th row of T . The upper estimate gives, for every set $S \subset \{1, \dots, n\}$ with $|S| \leq k$,

$$\left(\sum_{j \in S} r_{ij}^2 \right)^{1/2} \leq C.$$

Fix a set S with $|S| = k$ and let $(\varepsilon_j)_{j \in S}$ be independent Rademacher signs. For row i , set

$$Z_i = \frac{1}{\sqrt{k}} \sum_{j \in S} \varepsilon_j T_{ij}.$$

Hoeffding's inequality and the preceding row bound give

$$\mathbb{P}\{|Z_i| \geq 1\} \leq 2 \exp\left(-\frac{k}{2C^2}\right).$$

After a union bound over the n rows,

$$\mathbb{P}\left\{\max_i |Z_i| \geq 1\right\} \leq 2n \exp\left(-\frac{k}{2C^2}\right).$$

If $k > 2C^2 \log(2n)$, this probability is less than 1, so there is a sign choice with $\max_i |Z_i| < 1$. The vector $a_j = \varepsilon_j / \sqrt{k}$ on S and $a_j = 0$ off S is k -sparse and has $\|a\|_2 = 1$, but $\|Ta\|_\infty < 1$, contradicting the lower estimate.

How the counterexample supersedes it. The partial theorem uses random signs on the whole available support k , so it only contradicts the lower estimate when k is larger than a logarithmic threshold. The counterexample proof instead tests an intermediate support $h \leq \min\{k/(64C^2), \sqrt{\log n}\}$ and counts all signed h -sets. Since such an h still tends to infinity whenever $k \rightarrow \infty$, the remaining logarithmic window also collapses.

Human-review recommendation

Check the counting lemma and the high-coordinate extraction step. If those are correct, this is a complete negative answer to Problem 4 as stated.

References

- [1] Daniel J. Fresen, *Euclidean arrangements in Banach spaces*, arXiv:1309.5526, 2013.